

Gamma Functions over the Surreals

A sharp convexity classification, flat nonuniqueness,
and surcomplex phase domains

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Abstract

The classical Bohr–Møllerup characterization does not merely lose uniqueness on the surreal positive half-line: uniqueness can fail while normalization, strict log-convexity, every finite Gauss multiplication formula, all normalized finite-translation germs, and every Stirling coefficient are retained. We prove this by an explicit family and an exact coefficient classification. Let L_0 be the Taylor–Stirling extension of $\log \Gamma$, let $M(x)$ be the leading Conway monomial of a positive infinite surreal x , and let $\text{pp}(x)$ be its purely infinite part. For a class assignment $a : \mathfrak{M}_\infty \rightarrow \mathbf{No}$, set $h_a(x) = 0$ at positive finite arguments and $h_a(x) = a(M(x)) \text{pp}(x)$ at infinite ones. Then

$$L_0 + h_a \text{ is convex} \iff m a(m) \text{ is finite for every } m \in \mathfrak{M}_\infty.$$

In the affirmative case it is strictly convex. The proof uses exact tangent inequalities across all relative scales, not a mean-value theorem or the insufficient assertion that a second derivative is positive. The choice $a(m) = \lambda e^{-m}$ gives nonzero perturbations invisible to all algebraic orders at infinity. We also construct surcomplex log-Gamma extensions and identify their exact finite-phase exponentiation domain inside a finite-imaginary-part tube: at infinite real part x , the condition is $y \log x$ finite. Finally, compatibility with the Berarducci–Mantova scalar derivation eliminates every infinitesimal locally constant gauge in the stated category.

Status and provenance. The Taylor–Stirling construction, classical Gamma identities, restricted analytic transfer, and the Berarducci–Mantova derivation are established inputs. The sharp gauge classification and its combined consequences are proposed contributions proved here. No claim is made to have resolved a named published open problem. Their absence from all published literature is not certified; the targeted literature and repository audit is recorded in section 11. This draft has not been independently refereed or verified by a proof assistant. The accompanying exact symbolic checks test finite algebraic identities, not the full surreal theorems.

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1 The question and the results

1.1 A classical characterization confronted with new scales

For positive real arguments, the Gamma function is uniquely characterized by positivity, the normalization $\Gamma(1) = 1$, the recurrence $\Gamma(x+1) = x\Gamma(x)$, and convexity of $\log \Gamma$. This is the Bohr–Mollerup theorem [7, §5.5(iv)]. We investigate its literal surreal analogue, with convexity required at *all* positive surreal arguments and all surreal convex-combination parameters.

The issue is not whether Gamma can be extended. Costin and Ehrlich explicitly include Gamma and log-Gamma in their extension theory, using Taylor expansions at finite arguments and transseries at infinite ones [5, §§8,10.1(v)]. Rather, the question here is what additional information determines an extension, once the field contains infinitely separated scales. Does genuine global log-convexity remove the freedom left by a difference equation? Does imposing all finite multiplication identities help? Can agreement to every asymptotic order still conceal a nonzero discrepancy?

The results below give a sharp negative answer within a concrete and quite large family. Their force comes from the simultaneous requirements. A generic periodic multiplier would usually destroy convexity or multiplication identities; a generic infinitesimal change would usually spoil local germs. The construction here avoids all those failures, and its convexity boundary is proved in both directions.

1.2 Overview of the main theorems

Every surreal number has a Conway normal form. For positive infinite x , write $M(x)$ for its leading monomial with its real coefficient removed, and $\text{pp}(x)$ for the sum of its purely infinite terms. Precise conventions appear in section 2. The baseline L_0 is defined in section 3. Given any class assignment a on the positive infinite monomials, define

$$h_a(x) = \begin{cases} 0, & 0 < x \in \mathcal{O}, \\ a(M(x))\text{pp}(x), & x > \mathbb{R}, \end{cases} \quad L_a = L_0 + h_a, \quad \Gamma_a = \exp L_a. \quad (1.1)$$

Here $x > \mathbb{R}$ means that x exceeds every real number.

Theorem 1.1 (Sharp convexity and simultaneous invariance). *For the family (1.1), the following statements hold.*

- (i) L_a is convex on $\mathbf{No}_{>0}$ if and only if $ma(m) \in \mathcal{O}$ for every positive infinite monomial m . Under this condition L_a is strictly convex.
- (ii) For every a , independently of the convexity condition, Γ_a agrees with Γ_0 at every positive finite argument, satisfies $\Gamma_a(1) = 1$ and $\Gamma_a(x+1) = x\Gamma_a(x)$, and obeys

$$\Gamma_a(nx) = (2\pi)^{(1-n)/2} n^{nx-1/2} \prod_{k=0}^{n-1} \Gamma_a(x + k/n) \quad (1.2)$$

for every ordinary integer $n \geq 1$ and every $x > 0$.

- (iii) Whenever u is finite and $x, x+u > 0$,

$$\frac{\Gamma_a(x+u)}{\Gamma_a(x)} = \frac{\Gamma_0(x+u)}{\Gamma_0(x)}. \quad (1.3)$$

Every positive-order fine derivative of L_a equals that of L_0 .

(iv) The condition $|h_a(x)| \prec x^{-N}$ for every infinite $x > 0$ and every $N \in \mathbb{N}$ is equivalent to $|a(m)| \prec m^{-N}$ for every $m \in \mathfrak{M}_\infty$ and every $N \in \mathbb{N}$.

The notation $b \prec c$, for $c > 0$, means that b/c is infinitesimal. In particular, part (iv) is stronger than equality of a finite number of asymptotic terms. It supplies all of them at once. Theorem 1.1 is proved in sections 5 and 6, using the global tangent gap estimates of section 4.

Corollary 1.2 (Explicit strengthened nonuniqueness). *For each real λ , set*

$$a_\lambda(m) = \lambda e^{-m}. \quad (1.4)$$

The resulting functions Γ_λ are pairwise distinct and satisfy all the properties of Theorem 1.1, including strict log-convexity and flatness of their logarithmic and relative differences to every algebraic order. At $x = \omega$,

$$\log \Gamma_\lambda(\omega) - \log \Gamma_0(\omega) = \lambda \omega e^{-\omega}. \quad (1.5)$$

Thus normalization, recurrence, strict log-convexity, all finite Gauss formulas, the finite Taylor extension, all normalized finite-translation germs, and the full Stirling asymptotic expansion do not jointly characterize a surreal Gamma function.

The surcomplex result concerns actual functions, but does not assume a canonical exponential for arbitrary infinite imaginary arguments. On

$$\mathcal{T} = \{x + iy : x > 0, y \in \mathcal{O}\},$$

we construct locally Hahn-analytic functions $\mathcal{L}_a$ extending L_a . Their canonical exponentiation using finite angles is available exactly on

$$\Omega = \{x + iy \in \mathcal{T} : x \in \mathcal{O}\} \cup \{x + iy \in \mathcal{T} : x > \mathbb{R}, y \log x \in \mathcal{O}\}. \quad (1.6)$$

This exact assertion and its limits are proved in section 8. A different sort of compatibility restores rigidity: section 9 shows that the Berarducci–Mantova scalar chain rule forces any infinitesimal locally constant log-Gamma gauge to vanish.

1.3 Why the main result is not a transfer tautology

All finite real identities have suitable local Taylor extensions, but the main theorem compares arbitrary infinite arguments, including arguments of different leading monomials and arguments whose relative separation is smaller than every prescribed inverse power. The coefficient assignment a can vary independently on different monomials. Neither its existence nor the sharp bound $a(m) = O(m^{-1})$ follows by replacing real variables with surreal ones in a first-order formula about Gamma.

The new work is the compatibility between this freedom and *global* convexity. A small second derivative, or the equality $L''_a = L''_0 > 0$, cannot supply that compatibility by itself. The proof measures the convexity margin at every scale and proves that it dominates precisely the allowed gauges.

2 Normal forms, summability, and derivative conventions

2.1 Class and set conventions

We work in a classical theory of sets and classes such as NBG. The surreal numbers form a proper-class ordered field, and $\mathbf{No}[i] = \mathbf{No}[i]$ is its complexification. Each individual normal form and every family being summed here has set-sized support. A “class assignment” a is a class function; no set of all class functions is formed. Explicit choices such as (1.4) are definable without selecting one object from each member of a proper-class family.

Every particular finite calculation, together with the supports of its operands and the countably many series used in it, can be carried out in a set-sized Hahn workspace. Such a workspace need not be invariant under every operation discussed in the paper. When restricted analytic transfer is used, one may instead choose an adequately large $\mathbf{No}(\varepsilon)$ with ε an ordinal epsilon-number. Here $\mathbf{No}(\varepsilon)$ consists of the numbers of birthday less than ε ; equipped with the Taylor extensions of restricted analytic functions and the surreal exponential, it is an elementary extension of the corresponding real structure [4, Prop. 4.7]; see also [5, Prop. 93]. The proofs do not regard a proper class as an ordinary topological space in ZFC, and do not invoke a proper-class sum.

2.2 Finite parts and monomial scales

The standard normal-form background is due to Conway and Gonshor [1, 2]. In our notation,

$$x = \sum_{\alpha \in S} r_\alpha \omega^\alpha, \quad r_\alpha \in \mathbb{R} \setminus \{0\}, \quad (2.1)$$

where S is a reverse well-ordered set of surreal exponents. We use $\text{supp}(x) = \{\omega^\alpha : \alpha \in S\}$ for the *monomial* support, with $\text{supp}(0) = \emptyset$; S itself is the exponent support. In increasing-exponent formal Hahn notation $t^\gamma := \omega^{-\gamma}$, the term ω^α has exponent $-\alpha$. This is monomial notation, not exponentiation by a varying surreal exponent. We write

$$\mathcal{O} = \{x : |x| < n \text{ for some } n \in \mathbb{N}_{>0}\}, \quad \mathbf{m} = \{x : |x| < 1/n \text{ for all } n \in \mathbb{N}_{>0}\}.$$

Every $x \in \mathcal{O}$ has a unique standard part $\text{st}(x) \in \mathbb{R}$ with $x - \text{st}(x) \in \mathbf{m}$. Define

$$\text{pp}(x) = \sum_{\alpha > 0} r_\alpha \omega^\alpha, \quad \mathbb{J} = \{x : \text{supp}(x) \subseteq \{\omega^\alpha : \alpha > 0\}\}.$$

Then $\text{pp} : \mathbf{No} \rightarrow \mathbb{J}$ is an $\mathbb{R}$ -linear projection, its kernel is $\mathcal{O}$, and $x - \text{pp}(x)$ is finite. In particular,

$$\text{pp}(x + u) = \text{pp}(x) \quad (u \in \mathcal{O}). \quad (2.2)$$

This is an exact equality, not an asymptotic estimate.

For $x > \mathbb{R}$, define $M(x) = \omega^{\max S}$. Its leading real coefficient is positive, so $x/M(x)$ is finite with strictly positive standard part. Set

$$\mathfrak{M}_\infty = \{\omega^\alpha : \alpha > 0\}.$$

For positive infinite x and finite u ,

$$M(x + u) = M(x), \quad M(nx) = M(x), \quad \text{pp}(nx) = n \text{pp}(x) \quad (n \in \mathbb{N}_{>0}). \quad (2.3)$$

The same scaling assertions hold for any positive real scalar. Distinct monomials have different Archimedean classes: their ratio is infinite or infinitesimal, not a finite appreciable number.

For arbitrary surreal quantities, $f = O(g)$ means that f/g is finite, and $f \prec g$ means that f/g is infinitesimal; we apply these notations with $g > 0$ or to absolute values. We write $f \sim g$ when $f/g = 1 + \epsilon$ with $\epsilon \in \mathfrak{m}$. In estimates containing several variables, all assertions are pointwise for the indicated surreal inputs, not statements about an unspecified sequential limit.

2.3 Strong sums, not topological limits

A family of Hahn series is strongly summable when the union of its monomial supports is reverse well ordered and each monomial occurs in only finitely many members. Its sum is then formed coefficientwise. The positive-support summability lemma implies the following standard fact [3, 4]: if $\epsilon_1, \dots, \epsilon_d$ are infinitesimal, any formal power series with real or complex coefficients can be evaluated at these arguments as a strong sum. Formal sums, products, and substitutions with zero constant term then commute with evaluation.

For clarity, the reason is a support statement. Reverse the order on the exponent group, so the supports of the infinitesimals are well-ordered positive sets. The additive monoid generated by their finite union is well ordered, and a fixed exponent has only finitely many representations by finite words from those positive supports. Consequently each coefficient in a substituted formal series receives only finitely many contributions. This also justifies the two-variable expansions used below. No analytic radius of convergence of the real coefficient sequence is required.

In particular, factorially growing Bernoulli coefficients cause no obstacle to evaluating the Stirling series at x^{-1} for positive infinite x . This fact does *not* turn its ordinary partial sums into a convergent sequence in the fine surreal order topology. Series evaluated at surreal or surcomplex infinitesimals below have this strong, coefficientwise meaning. Ordinary real or complex analytic series and formal series in indeterminates are identified separately where they are used.

2.4 Fine coordinate derivatives versus a scalar derivation

A fine derivative is the usual epsilon–delta derivative with surreal positive tolerances and surreal increments. For a class function f on an open interval, $f'(x) = d$ means that for every $\epsilon > 0$ there is a $\delta > 0$ such that

$$0 < |u| < \delta \implies \left| \frac{f(x+u) - f(x)}{u} - d \right| < \epsilon.$$

The complex version uses the ordered modulus on $\mathbf{No}[i]$. Class quantifiers are interpreted in the ambient class theory. All derivative assertions below are established on explicit local neighborhoods.

A function constant on cosets modulo $\mathcal{O}$ is finely locally constant: the neighborhood $|u| < 1$ stays in the same coset. Every fine derivative of such a function is zero, although its values on different cosets may change. This elementary observation is not the main theorem, but it explains why a proof of global convexity must use more than local derivatives.

In section 9 the symbol ∂ denotes instead the Berarducci–Mantova derivation *of the field of scalar surreal numbers*. It does not differentiate a class function by varying its argument. The two operations are related only when a chain-rule compatibility statement is proved or explicitly imposed.

3 The Taylor–Stirling log-Gamma function

3.1 Definition on the three positive regimes

Let $\ell(r) = \log \Gamma(r)$ for real $r > 0$, and let γ be Euler’s constant. Put

$$c = \frac{1}{2} \log(2\pi), \quad b_j = \frac{B_{2j}}{2j(2j-1)} \quad (j \geq 1), \quad (3.1)$$

where the Bernoulli numbers have generating function $t/(e^t - 1) = \sum_{j \geq 0} B_j t^j / j!$. Define $L_0 : \mathbf{No}_{>0} \rightarrow \mathbf{No}$ by

$$L_0(r + \epsilon) = \sum_{k \geq 0} \frac{\ell^{(k)}(r)}{k!} \epsilon^k \quad (r \in \mathbb{R}_{>0}, \epsilon \in \mathbf{m}), \quad (3.2)$$

by

$$L_0(x) = -\log x - \gamma x + \sum_{k \geq 2} \frac{(-1)^k \zeta(k)}{k} x^k \quad (0 < x \in \mathbf{m}), \quad (3.3)$$

and by

$$L_0(x) = \left(x - \frac{1}{2}\right) \log x - x + c + R(x), \quad R(x) = \sum_{j \geq 1} b_j x^{-(2j-1)} \quad (x > \mathbb{R}). \quad (3.4)$$

These regimes partition the positive surreal half-line. The coefficients and normalizations in (3.3) and (3.4) are the classical ones [8, 10, §§5.7, 5.11]. We define $\Gamma_0(x) = \exp L_0(x)$ using the usual surreal exponential.

Proposition 3.1 (Well-definedness and local calculus). *All three definitions are well-defined strong sums. The function L_0 is finely differentiable to every finite order. On each of the three regimes, its derivatives are obtained by the displayed Taylor or formal Stirling rules, together with $(\log x)' = 1/x$. In particular, for infinite $x > 0$,*

$$L'_0(x) = \log x - \frac{1}{2x} + R'(x), \quad (3.5)$$

$$R(x) = \frac{1}{12x} + O(x^{-3}), \quad R'(x) = -\frac{1}{12x^2} + O(x^{-4}), \quad (3.6)$$

$$L''_0(x) = \frac{1}{x} + \frac{1}{2x^2} + \frac{1}{6x^3} + O(x^{-5}). \quad (3.7)$$

For positive infinitesimal x ,

$$L_0(x) = -\log x + O(x), \quad L'_0(x) = -x^{-1} + O(1). \quad (3.8)$$

Proof. The summability statement follows by substituting ϵ , x , or x^{-1} , respectively, into a formal power series with real coefficients. All three regimes are fine-open: a sufficiently small increment leaves a point finite appreciable, positive infinitesimal, or infinite, respectively.

Here is also the needed derivative justification, rather than just a formal notation. In the finite appreciable case the Taylor expansion can be re-expanded jointly in ϵ and an infinitesimal increment u . The formal difference after subtracting its linear term is u^2 times a strongly evaluable series; on a smaller local neighborhood that series has a fixed finite bound. The epsilon–delta derivative follows by choosing $|u|$ below the requested tolerance divided by that bound. Higher finite derivatives are obtained by repeating this argument.

For (3.3), take $|u| < x/2$ and expand $\log(x+u) = \log x + \log(1+u/x)$, then shrink to make u/x infinitesimal. For (3.4), use $(x+u)^{-k} = x^{-k}(1+u/x)^{-k}$ and the same logarithmic identity. The coefficients of the remainder may now contain fixed powers of x or x^{-1} , but are still bounded by a fixed surreal quantity on a sufficiently small neighborhood. An epsilon-delta choice is allowed to depend on that quantity. The joint power series in x^{-1} and u/x is strongly summable, so regrouping is legitimate. This proves the fine derivative and its iterates, not just a coefficientwise derivative in an unrelated ring. The explicit estimates follow from the first nonzero terms. $\square$

3.2 An exact recurrence from a formal identity

Introduce the formal log-Laurent expression

$$S(X) = \left(X - \frac{1}{2}\right) \log X - X + c + \sum_{j \geq 1} b_j X^{-(2j-1)}.$$

Translations of $\log X$ are expanded by $\log(X+t) = \log X + \log(1+t/X)$ for real t .

Lemma 3.2 (Formal shift identity). *As a formal expression at $X = +\infty$,*

$$S(X+1) - S(X) = \log X. \quad (3.9)$$

Consequently,

$$L_0(x+1) - L_0(x) = \log x \quad (x > 0). \quad (3.10)$$

Proof. One rigorous derivation of (3.9) uses only classical asymptotic algebra. The real function ℓ has the complete Stirling expansion S and satisfies $\ell(X+1) - \ell(X) = \log X$ [7, 10]. To determine any fixed inverse-power coefficient of the difference, retain sufficiently many terms in the two real expansions and expand the shift by the binomial and logarithmic formulas. The remainder is of strictly smaller order than the coefficient under examination. Uniqueness of real asymptotic coefficients forces that coefficient to vanish. This holds at every finite order and proves the formal identity. Equivalently, it follows directly from the Bernoulli generating function; the supplied symbolic program checks the cancellation through a substantial finite order.

For infinite surreal x , the difference in (3.9) is an identity of formal power series in x^{-1} after the elementary terms have been removed. Strong evaluation therefore gives an exact equality, with no unaccounted remainder. At finite appreciable x , use the local Taylor identity for the ordinary recurrence. At positive infinitesimal x , formula (3.3) is exactly the Taylor series for $\ell(1+x) - \log x$. These cases prove (3.10) everywhere. $\square$

3.3 All finite multiplication formulas

Lemma 3.3 (No nonconstant periodic inverse-power series). *Let $F \in X^{-1}\mathbb{R}[[X^{-1}]]$ and let $t \in \mathbb{R} \setminus \{0\}$. If $F(X+t) = F(X)$ formally, then $F = 0$.*

Proof. A nonzero F has a first term dX^{-k} with $d \neq 0$ and $k \geq 1$. The first term of $F(X+t) - F(X)$ is $-ktdX^{-k-1}$, which is nonzero. $\square$

Proposition 3.4 (Exact Gauss identity for the baseline). *For every ordinary integer $n \geq 1$ and every positive surreal x ,*

$$L_0(nx) = \sum_{k=0}^{n-1} L_0(x + k/n) + (nx - \frac{1}{2}) \log n - (n-1)c. \quad (3.11)$$

Exponentiating gives (1.2) for $a = 0$.

Proof. For finite arguments, extend the classical identity [7, Eq. 5.5.6] by the local Taylor rules, using (3.3) at an infinitesimal argument. For the infinite case form the formal defect

$$F_n(X) = \sum_{k=0}^{n-1} S(X + k/n) - S(nX) + (nX - \tfrac{1}{2}) \log n - (n-1)c.$$

Expanding only the elementary leading terms shows that all terms involving $X \log X$, X , $\log X$, and the constant coefficient cancel; hence $F_n \in X^{-1}\mathbb{R}[[X^{-1}]]$. Under $X \mapsto X + 1/n$, the difference of the sum is $S(X+1) - S(X) = \log X$, the difference of $-S(nX)$ is $-\log(nX)$, and the remaining linear term contributes $\log n$. Thus $F_n(X + 1/n) = F_n(X)$. Lemma 3.3 gives $F_n = 0$, and strong evaluation gives (3.11). $\square$

Remark 3.5 (What a formal equality does and does not decide). The exact definition (3.4) chooses the value of one strongly evaluated formal series. Equality of *asymptotic coefficients* is a weaker condition: it permits additional terms smaller than every x^{-N} . The gauges below exploit this distinction. They do not contradict uniqueness of the strong sum of a specified family.

4 Global convexity from multiscale tangent gaps

4.1 A criterion that remains valid without completeness

For a differentiable function F on an interval in an ordered field, define its tangent gap by

$$D_F(y, x) = F(y) - F(x) - F'(x)(y - x). \quad (4.1)$$

We will prove positivity of this expression directly.

Lemma 4.1 (Tangent support and convexity). *Let I be an interval in an ordered field and let $F : I \rightarrow K$ be finely differentiable on its interior. If F is convex, then $D_F(y, x) \geq 0$ whenever x is interior and $y \in I$. Conversely, if $D_F(y, x) > 0$ for every two distinct points of an open interval, then F is strictly convex on that interval.*

Proof. For the first assertion, convexity gives

$$F(x + t(y - x)) - F(x) \leq t(F(y) - F(x)) \quad (0 < t < 1).$$

Dividing by t and using the derivative at x shows that $F'(x)(y - x) \leq F(y) - F(x)$. This step uses only the defining fine epsilon–delta inequality: a strictly negative proposed gap would contradict that inequality for a sufficiently small positive t . No sequential convergence or supremum is needed.

For the converse let $z = (1 - \theta)x + \theta y$ with $x \neq y$ and $0 < \theta < 1$. Apply the strict support inequality at z , once with endpoint x and once with endpoint y . Multiply by $1 - \theta$ and θ , respectively, and add. The derivative terms cancel, giving $F(z) < (1 - \theta)F(x) + \theta F(y)$. $\square$

Write $D_0 = D_{L_0}$. The next lemma is the analytic core of the article. Its relative-close case is especially important: the increment need not be an ordinary real-sized perturbation, or even a fixed inverse power of x .

Lemma 4.2 (Infinite-argument tangent gaps). *Suppose $x, y > \mathbb{R}$ and $x \neq y$. Put $q = y/x$. Then $D_0(y, x) > 0$. More precisely:*

(i) If $q = 1 + u$ with $0 \neq u \in \mathfrak{m}$, then

$$D_0(y, x) \sim \frac{(y - x)^2}{2x}. \quad (4.2)$$

(ii) If q is finite appreciable with $r = \text{st}(q) \neq 1$, then

$$D_0(y, x) \sim x(r \log r - r + 1). \quad (4.3)$$

(iii) If q is positive infinite, then

$$D_0(y, x) \sim y \log q. \quad (4.4)$$

(iv) If q is positive infinitesimal, then

$$D_0(y, x) \sim x. \quad (4.5)$$

Proof. Substituting (3.4) and (3.5) gives the exact identity

$$D_0(y, x) = x\Phi(q) + \frac{1}{2}\Psi(q) + E_R(y, x), \quad (4.6)$$

$$\Phi(q) = q \log q - q + 1, \quad \Psi(q) = q - 1 - \log q, \quad (4.7)$$

$$E_R(y, x) = R(y) - R(x) - R'(x)(y - x). \quad (4.8)$$

We estimate this expression in each regime.

Relative-close arguments. Let $q = 1 + u$ with $u \in \mathfrak{m} \setminus \{0\}$. The formal identities

$$\Phi(1 + u) = \frac{u^2}{2} - \frac{u^3}{6} + \frac{u^4}{12} - \cdots, \quad (4.9)$$

$$\Psi(1 + u) = \frac{u^2}{2} - \frac{u^3}{3} + \frac{u^4}{4} - \cdots \quad (4.10)$$

are strongly evaluable. Writing $k = 2j - 1$, the remainder is

$$E_R(y, x) = \sum_{j \geq 1} b_j x^{-k} ((1 + u)^{-k} - 1 + ku). \quad (4.11)$$

Each bracket is divisible by u^2 as a formal power series, with quadratic coefficient $k(k + 1)/2$. The joint substitution in x^{-1} and u is legitimate by positive-support summability. It gives $E_R(y, x) = O(x^{-1}u^2)$. Hence

$$D_0(y, x) = \frac{xu^2}{2}(1 + O(u) + O(x^{-1})),$$

where both error terms inside the parentheses are infinitesimal. This proves (4.2) even when u lies below every x^{-N} : there is no discarded error independent of u^2 .

Comparable but not relative-close arguments. If $r = \text{st}(q) > 0$ and $r \neq 1$, the finite Taylor extension of $\log$ gives $\Phi(q) = \Phi(r) + \epsilon$ with $\epsilon \in \mathfrak{m}$. The real number $\Phi(r)$ is strictly positive. The term $\Psi(q)$ is finite, and (3.6) gives $E_R(y, x) = O(x^{-1})$. Thus the first term of (4.6) dominates and yields (4.3).

An infinitely larger second argument. If $q > \mathbb{R}$, then $\log q > \mathbb{R}$ and $\Phi(q) \sim q \log q$. Also $\Psi(q) = O(q)$ and

$$E_R(y, x) = O(y^{-1}) + O(x^{-1}) + O(y/x^2).$$

After division by $y \log q$ all these error terms are infinitesimal, proving (4.4).

An infinitely smaller, but still infinite, second argument. For $q \in \mathfrak{m}_{>0}$, $q \log q \in \mathfrak{m}$, so $\Phi(q) \sim 1$. Since $y > 1$, we have $0 < \log(1/q) = \log(x/y) < \log x \prec x$. Thus $\Psi(q) \prec x$, while $E_R(y, x)$ is infinitesimal by (3.6). This proves (4.5).

The elementary exponential comparisons just used, such as $\log x \prec x$ at positive infinite x , hold for the usual surreal exponential; they also follow from the elementary exponential-field extension theorem [4]. The four cases exhaust all positive q . $\square$

Lemma 4.3 (Finite and mixed tangent gaps). *For two distinct positive finite surreals x, y , $D_0(y, x) > 0$. If exactly one of x, y is infinite, $D_0(y, x)$ is positive infinite.*

Proof. First suppose that both arguments are finite. Choose a positive ordinary integer N with $x, y < N$. The real function $A_N(t) = \log \Gamma(1 + t)$ is analytic on a neighborhood of $[0, N]$; its restriction can be encoded as a restricted analytic function after a real affine change of variables. On $0 < t < N$,

$$L_0(t) = A_N(t) - \log t,$$

where A_N is interpreted by its Taylor extension. The classical function $\log \Gamma$ is strictly convex, and its strict tangent inequality on $(0, N)$ is a first-order assertion involving A_N , its restricted analytic derivative, $\log$, and the field operations. The restricted analytic exponential-field transfer theorem of van den Dries and Ehrlich [4] transfers this assertion to our x, y . This use of transfer is on a fixed standard bounded interval; it does not assume a globally named Gamma function on $\mathbf{No}$.

Now let x be finite and y infinite. If x is appreciable then $L_0(x)$ and $L'_0(x)$ are finite, so (3.4) gives

$$D_0(y, x) \sim y \log y > \mathbb{R}.$$

If x is positive infinitesimal, (3.8) gives instead

$$D_0(y, x) \sim y \log y + y/x > \mathbb{R}.$$

Indeed the errors are sums of $O(y)$, $O(\log y)$, $O(|\log x|)$, and finite terms. They are negligible relative to the positive displayed sum: in particular $|\log x| \prec x^{-1} \prec y/x$. There is no assumption about the relative sizes of $y \log y$ and y/x .

Finally let x be infinite and y finite. The infinite expansion implies

$$xL'_0(x) - L_0(x) = x + O(\log x) \sim x, \quad yL'_0(x) = O(\log x) \prec x.$$

There is a fixed real constant C such that $\log \Gamma(t) > -C$ for every positive real t ; for example this follows from the positive continuous real Gamma function, its behavior at zero, and Stirling's expansion at infinity. The same bound holds for finite positive y , by the bounded-interval transfer argument above. Therefore

$$D_0(y, x) = L_0(y) + xL'_0(x) - L_0(x) - yL'_0(x) > x/2.$$

This remains true if $L_0(y)$ itself is positive infinite because y is extraordinarily small. All mixed cases are covered. $\square$

Theorem 4.4 (Global strict log-convexity of the baseline). *For all distinct $x, y > 0$,*

$$L_0(y) > L_0(x) + L'_0(x)(y - x). \quad (4.12)$$

Consequently L_0 is strictly convex on the entire surreal positive half-line, with arbitrary surreal convex-combination parameters.

Proof. Combine Lemmas 4.1 to 4.3. $\square$

4.2 The geometric meaning of the estimates

If $y = x + \Delta$ with $\Delta \prec x$, the baseline has a convexity margin of size Δ^2/x . If Δ is finite, a gauge constant under finite translations changes nothing. If Δ is infinite, a gauge increment of size at most Δ/x is infinitesimal relative to that margin. This is the source of the sharp coefficient threshold $a(M(x)) = O(x^{-1})$. At arguments with different leading scales the margin is already infinite, so any finite gauge discrepancy is harmless. These observations are made exact in the next section.

5 The sharp monomial-gauge classification

5.1 Algebraic invariance and local invisibility

For an arbitrary class assignment $a : \mathfrak{M}_\infty \rightarrow \mathbf{No}$, define h_a, L_a , and Γ_a by (1.1). No regularity of a is assumed.

Lemma 5.1 (Gauge identities). *For $x, x + u > 0$ and $u \in \mathcal{O}$,*

$$h_a(x + u) = h_a(x). \quad (5.1)$$

For every positive ordinary integer n ,

$$h_a(nx) = nh_a(x). \quad (5.2)$$

The function h_a is finely locally constant on $\mathbf{No}_{>0}$; in particular $h_a^{(k)} = 0$ for every finite $k \geq 1$.

Proof. A finite translation cannot change an infinite argument into a finite one or conversely. In the finite case both sides of (5.1) vanish. In the infinite case use (2.2) and (2.3). The same rules give (5.2); the finite case again vanishes. For local constancy restrict to $|u| < \min(1, x/2)$. Repeated differentiation of a locally constant function gives zero. $\square$

Proposition 5.2 (Identities shared by every gauge). *Every Γ_a is positive, agrees with Γ_0 at all positive finite arguments, is normalized at 1, and satisfies recurrence and every finite Gauss multiplication formula. It also satisfies (1.3), and $L_a^{(k)} = L_0^{(k)}$ for all finite $k \geq 1$.*

Proof. Positivity follows from the ordered exponential, and agreement and normalization follow from $h_a = 0$ on positive finite inputs. Equation (5.1) with $u = 1$ preserves (3.10). For multiplication,

$$\sum_{k=0}^{n-1} h_a(x + k/n) = nh_a(x) = h_a(nx),$$

which preserves (3.11). Exponentiate using $\exp(v+w) = \exp(v)\exp(w)$. For any finite u , subtracting $L_a(x)$ from $L_a(x+u)$ cancels the gauge exactly and proves the normalized ratio identity. The derivative assertion follows from Lemma 5.1. $\square$

Warning 5.3 (Which derivatives coincide). The equality just proved concerns the derivatives of *log-Gamma*. In general $\Gamma_a^{(k)}(x)$ does not equal $\Gamma_0^{(k)}(x)$; on a local neighborhood it equals $e^{h_a(x)}\Gamma_0^{(k)}(x)$. Consequently normalized derivatives such as $\Gamma_a^{(k)}/\Gamma_a$ agree, while unnormalized values need not.

5.2 Sufficiency of the finite coefficient bound

Theorem 5.4 (Convexity under the sharp bound). *If $ma(m) \in \mathcal{O}$ for every $m \in \mathfrak{M}_\infty$, then*

$$D_{L_a}(y, x) > 0 \quad (x, y > 0, x \neq y). \quad (5.3)$$

In particular, L_a is strictly convex.

Proof. For an infinite x with $m = M(x)$,

$$h_a(x) = (ma(m)) \frac{\text{pp}(x)}{m} \in \mathcal{O}.$$

Thus every gauge value is finite, although there need not be a single real bound working for all monomials. By local constancy,

$$D_{L_a}(y, x) = D_0(y, x) + h_a(y) - h_a(x). \quad (5.4)$$

If x, y are finite, the perturbation is zero and Lemma 4.3 applies. If exactly one is infinite, the baseline gap is positive infinite while the perturbation is finite, so the conclusion again follows.

Suppose both are infinite. If $M(x) \neq M(y)$, their ratio is infinite or infinitesimal. By Lemma 4.2, the baseline gap is positive infinite, and a finite perturbation cannot change its sign. If $M(x) = M(y) = m$ but $\text{st}(y/x) \neq 1$, the same conclusion follows from (4.3).

It remains to consider $M(x) = M(y) = m$ and $\Delta = y - x \prec x$ with $\Delta \neq 0$. If Δ is finite, (5.1) makes the perturbation exactly zero. If Δ is infinite, linearity of the purely infinite projection gives

$$h_a(y) - h_a(x) = a(m) \text{pp}(\Delta), \quad \text{pp}(\Delta) \sim \Delta.$$

By (4.2),

$$\frac{|h_a(y) - h_a(x)|}{D_0(y, x)} \sim \frac{2|a(m)|x}{|\Delta|} \in \mathfrak{m},$$

unless $a(m) = 0$, in which case the numerator vanishes outright. Here $a(m)x$ is finite because x/m and $ma(m)$ are finite, whereas $|\Delta|$ is infinite. Thus (5.4) is strictly positive in the last case as well. Lemma 4.1 proves strict convexity. $\square$

5.3 Necessity and an explicit violating tangent

Theorem 5.5 (Necessity of the finite coefficient bound). *If $ma(m)$ is infinite in absolute value for some $m \in \mathfrak{M}_\infty$, then L_a is not convex. In fact a violating tangent can be found using two positive infinite arguments having the same leading monomial m .*

Proof. Fix such an m and abbreviate $a = a(m) \neq 0$. Both m and $|a|m$ are positive infinite. Let

$$b = \min(m, |a|m), \quad \delta = \text{pp}(\sqrt{b}).$$

Then δ is positive purely infinite, $\delta \sim \sqrt{b}$, and

$$\delta \prec m, \quad \delta \prec |a|m. \quad (5.5)$$

Set $x = m$ and $y = m - \text{sgn}(a)\delta$. These arguments are positive infinite, $M(y) = m$, and y is purely infinite. Therefore

$$h_a(y) - h_a(m) = -|a|\delta.$$

On the other hand, (4.2) gives

$$D_0(y, m) \sim \frac{\delta^2}{2m} \prec |a|\delta,$$

since the ratio is asymptotic to $\delta/(2|a|m)$. Using $L'_a = L'_0$ in (5.4), we obtain $D_{L_a}(y, m) < 0$. Convex differentiable functions have nonnegative tangent gaps by Lemma 4.1. Thus L_a is not convex. $\square$

Corollary 5.6 (The exact coefficient region). *Within the entire monomialwise family (1.1),*

$$L_a \text{ is convex} \iff \forall m \in \mathfrak{M}_\infty, a(m) = O(m^{-1}).$$

There are no merely convex but non-strictly-convex members of this family.

The coefficient condition is pointwise in m . Arbitrarily different finite bounds at different monomials are allowed. Both positive and negative gauges are allowed. This is not a classification of *all* surreal Gamma extensions or of all possible periodic gauges; it is an exact classification of the explicitly stated large family.

5.4 Examples at and beyond the boundary

Example 5.7 (A non-flat boundary gauge). For a real λ , take $a(m) = \lambda/m$. Then

$$h_a(x) = \lambda \frac{\text{pp}(x)}{M(x)}$$

is finite at infinite x , and the theorem proves strict log-convexity. At $x = m$ it equals λ , so unless $\lambda = 0$ this gauge changes the constant order of the logarithmic asymptotics. It demonstrates that the allowable convexity region is much larger than the flat subfamily.

Example 5.8 (A gauge that looks locally harmless but is not convex). Choose one infinite monomial m_0 , put $a(m_0) = 1$, and put $a(m) = 0$ at all other monomials. Every positive-order fine derivative of L_a still coincides with that of L_0 . Nevertheless, choosing $x = m_0$, $y = m_0 - \sqrt{m_0}$ gives

$$D_0(y, x) \sim \frac{1}{2}, \quad h_a(y) - h_a(x) = -\sqrt{m_0},$$

so the tangent gap is negative. This is a concrete failure of the implication “positive second derivative implies convexity” for the fine calculus at hand. It also independently checks the orientation and size of the necessity construction.

6 Flat families and the strengthened failure of uniqueness

6.1 An exact flatness criterion

Definition 6.1. An assignment $a : \mathfrak{M}_\infty \rightarrow \mathbf{No}$ is *algebraically flat* if

$$|a(m)| \prec m^{-N} \quad (m \in \mathfrak{M}_\infty, N \in \mathbb{N}).$$

A function h on $\mathbf{No}_{>0}$ is algebraically flat at infinite arguments if $|h(x)| \prec x^{-N}$ for every $x > \mathbb{R}$ and every ordinary $N \in \mathbb{N}$.

Proposition 6.2 (Coefficient flatness equals function flatness). *For the family h_a , coefficient flatness and function flatness are equivalent. Under either condition L_a is strictly convex and*

$$\frac{\Gamma_a(x)}{\Gamma_0(x)} - 1 \prec x^{-N} \quad (x > \mathbb{R}, N \in \mathbb{N}). \quad (6.1)$$

Proof. Let $m = M(x)$. Since x/m is finite appreciable and $\text{pp}(x)/m$ is finite, the estimate $|a(m)| \prec m^{-(N+1)}$ gives $|a(m) \text{pp}(x)| \prec x^{-N}$. This proves function flatness. Conversely, evaluate at $x = m$, where $h_a(m) = ma(m)$, and use the function estimate at order N to obtain a coefficient estimate at order $N + 1$. Together with the weaker orders this is coefficient flatness.

In particular $ma(m)$ is infinitesimal, hence finite, so Theorem 5.4 applies. Finally $h_a(x)$ is infinitesimal and $\exp h_a(x) - 1 \sim h_a(x)$ when it is nonzero, by its strongly summable Taylor expansion. This gives (6.1); the zero case is immediate. $\square$

Parts (i)–(iii) of Theorem 1.1 follow from Theorems 5.4 and 5.5 and Proposition 5.2, and part (iv) follows from Proposition 6.2. This completes its proof.

Warning 6.3 (Relative flatness does not bound an absolute Gamma difference). The flat quantities are $L_a - L_0$ and $\Gamma_a/\Gamma_0 - 1$. Multiplication by Γ_0 need not preserve flatness. For example, with $a_1(m) = e^{-m}$, put $h = \omega e^{-\omega}$. Then

$$\Delta := \Gamma_1(\omega) - \Gamma_0(\omega) = \Gamma_0(\omega)(e^h - 1) \sim \Gamma_0(\omega)\omega e^{-\omega} > 0.$$

Since $(e^h - 1)/h = 1 + \eta$ for an infinitesimal η , taking logarithms gives

$$\log \Delta = \left(\omega + \frac{1}{2}\right) \log \omega - 2\omega + c + R(\omega) + \log(1 + \eta) \sim \omega \log \omega.$$

Thus $\log \Delta > N \log \omega$, and hence $\Delta > \omega^N$, for every ordinary $N \in \mathbb{N}$. The absolute difference can therefore be enormous despite agreement to every relative algebraic order.

6.2 Explicit families with arbitrarily many parameters

Proof of Corollary 1.2. For every positive infinite m and ordinary N , $m^N e^{-m} \in \mathfrak{m}$. To see this directly, the real exponential eventually exceeds every prescribed positive real multiple of t^{N+1} ; the same inequality holds at infinite surreal t by exponential-field transfer [4]. Consequently $a_\lambda(m) = \lambda e^{-m}$ is flat for real λ . The main theorem gives all the asserted identities and strict convexity. At $x = \omega$, $M(x) = \text{pp}(x) = \omega$, yielding (1.5). Distinct real λ give distinct logarithms there, and injectivity of the surreal exponential gives distinct Gamma functions. $\square$

Theorem 6.4 (Independent monomial parameters). *Let $S \subseteq \mathfrak{M}_\infty$ be any set. To each function $\lambda : S \rightarrow \mathbb{R}$ associate the definable class assignment*

$$a_\lambda(m) = \begin{cases} \lambda(m)e^{-m}, & m \in S, \\ 0, & m \notin S. \end{cases}$$

The map $\lambda \mapsto \Gamma_{a_\lambda}$ is injective, and all its members have the properties of Corollary 1.2.

Proof. Flatness is pointwise in m , so no boundedness of the real parameters as m varies is needed. Distinct parameter functions differ at some $m \in S$, where their log-Gamma values differ by the nonzero quantity $(\lambda(m) - \mu(m))me^{-m}$. $\square$

Formally, this indexed family is the single class function $(\lambda, x) \mapsto \Gamma_{a_\lambda}(x)$ on $\mathbb{R}^S \times \mathbf{No}_{>0}$. “Injective” means that different indices are separated by some argument; it does not make proper classes into values of a set function. This gives independent parameters on sets of any size available in the ambient set theory, without forming a set of all class functions.

Example 6.5 (The same local data at visibly different arguments). For the one-parameter family and $\lambda \neq 0$,

$$\begin{aligned} h_\lambda(\omega) &= \lambda\omega e^{-\omega}, \\ h_\lambda(\omega + 17 + \omega^{-1}) &= \lambda\omega e^{-\omega}, \\ h_\lambda(2\omega) &= 2\lambda\omega e^{-\omega}, \\ h_\lambda(\omega + \sqrt{\omega}) &= \lambda(\omega + \sqrt{\omega})e^{-\omega}, \\ h_\lambda(\omega^2) &= \lambda\omega^2 e^{-\omega^2}. \end{aligned}$$

The fourth line changes the gauge because the increment is infinite, although it is infinitesimal relative to the argument. The second line does not change it because the entire increment is finite. In the third line the exponent is still $-\omega$, not -2ω : $M(2\omega) = \omega$. Confusing a leading monomial with a leading term would invalidate the multiplication identities.

6.3 Even the usual signed remainder bounds survive

Let

$$S_N(x) = \left(x - \frac{1}{2}\right) \log x - x + c + \sum_{j=1}^N b_j x^{-(2j-1)} \quad (N \in \mathbb{N}). \quad (6.2)$$

An empty sum is zero. The following strengthens equality of asymptotic coefficients by retaining the direction and size of each ordinary-order Stirling remainder.

Theorem 6.6 (Enveloping inequalities are still non-unique). *For every algebraically flat a , every $x > 0$, and every ordinary $N \in \mathbb{N}$,*

$$0 < (-1)^N (L_a(x) - S_N(x)) < |b_{N+1}| x^{-(2N+1)}. \quad (6.3)$$

Thus adding all these inequalities to the requirements in Corollary 1.2 does not restore uniqueness.

Proof. For finite positive x , $L_a = L_0$ and the classical strict real remainder bounds for log-Gamma [10, §5.11(ii)] transfer on a standard bounded interval, exactly as in Lemma 4.3. The elementary terms in S_N introduce only powers and logarithms. For completeness, strictness follows directly from Binet’s real formula [9, Eq. 5.9.10]:

$$\ell(x) - S_0(x) = 2 \int_0^\infty \frac{\arctan(t/x)}{e^{2\pi t} - 1} dt \quad (x \in \mathbb{R}_{>0}).$$

The finite expansion

$$\arctan v = \sum_{k=0}^{N-1} \frac{(-1)^k v^{2k+1}}{2k+1} + (-1)^N \int_0^v \frac{u^{2N}}{1+u^2} du$$

has a signed remainder strictly between 0 and $v^{2N+1}/(2N+1)$ for $v > 0$. Integrating against the positive Binet kernel gives the strict first-omitted-term bounds; the integrated polynomial

coefficients are the b_j in (3.1). This is an ordinary real integral argument, used only before transfer, not a surreal integral.

For infinite x , the coefficients b_j alternate in sign, with $\operatorname{sgn} b_j = (-1)^{j+1}$. Therefore the strong-series remainder has the form

$$(-1)^N(L_0(x) - S_N(x)) = |b_{N+1}|x^{-(2N+1)} - |b_{N+2}|x^{-(2N+3)} + O(x^{-(2N+5)}).$$

Flatness makes $h_a(x)$ infinitesimal relative to the second displayed term. The remainder itself is positive because of its first term, and the difference from the upper bound is positive because of the second term. This proves both strict inequalities after adding the gauge. $\square$

6.4 What “all orders” means here

All differentiation orders, multiplication indices, and Stirling truncation indices in this article are ordinary nonnegative integers. A transfinite product, a hyperinteger-indexed recurrence scheme, or a prescription for summing a proper-class expansion would be a different additional structure. None is smuggled into (1.2) or (6.3).

Likewise, flatness here does not mean zero. At an infinite x the ordered field contains nonzero elements smaller than every ordinary inverse power of x . The explicit examples use that room. Requiring equality to a chosen full strong sum would remove it by definition, but would be a stronger normalization than the classical asymptotic expansion and its usual finite-order error inequalities.

7 Local Gamma calculus and the failure of classical propagation

7.1 Normalized germs and polygamma identities

For a positive argument x , the local multiplicative germ of Γ_a normalized at x is

$$u \mapsto \frac{\Gamma_a(x+u)}{\Gamma_a(x)}.$$

It is exactly the baseline germ. This includes finite translations which are not infinitesimal, as long as positivity of the shifted argument is retained. In particular, all logarithmic derivatives and all normalized finite-order derivatives coincide.

Writing $\psi = L'_0$ gives, by differentiating the already established identities,

$$\psi(x+1) - \psi(x) = x^{-1}, \tag{7.1}$$

$$n\psi(nx) = \sum_{k=0}^{n-1} \psi(x + k/n) + n \log n. \tag{7.2}$$

Higher fine derivatives give their polygamma counterparts. At positive infinite x ,

$$\psi(x) = \log x - \frac{1}{2x} - \frac{1}{12x^2} + \frac{1}{120x^4} - \frac{1}{252x^6} + \cdots,$$

with strong summation, and

$$(-1)^{r+1}\psi^{(r)}(x) \sim (r-1)!x^{-r} > 0 \quad (r \geq 1).$$

At finite positive arguments the same sign is obtained from the ordinary polygamma functions by restricted analytic transfer. These facts agree with the classical formulas [11], but do not distinguish any two members of our family.

Warning 7.1 (An illegitimate substitution into an ordinary sum). For real $x > 0$, the familiar identity $\psi'(x) = \sum_{k \geq 0} (x+k)^{-2}$ is an ordinarily convergent sum. At infinite surreal x , its summands all have the same leading monomial $M(x)^{-2}$, with nonzero real leading coefficient. The family is therefore *not* strongly summable: that monomial occurs infinitely often. The correct expression used here is the differentiated Stirling strong series, not that ordinary series with an infinite number substituted into it.

7.2 Why the real uniqueness argument does not transfer wholesale

The unit recurrence propagates values along $x + \mathbb{Z}$, and finite multiplication formulas relate a finite number of comparable arguments. Neither operation moves an infinite argument to the real interval $(0, 1]$. More strongly, our normalized germs already prescribe every finite translation from every starting point, and the gauge still survives.

Real Bohr–Møllerup arguments eventually use the Archimedean property to squeeze a difference to zero through real error bounds. In the present field, an error smaller than x^{-N} for every ordinary N can be nonzero. Theorem 6.6 makes the limitation explicit even for signed Stirling inequalities. The failure is not a defect in the classical theorem: its ambient ordered field and its globally available means of propagation have changed.

There is no contradiction with elementary extension of the restricted analytic exponential structure. That language names analytic functions on fixed real compact boxes and the exponential. The normal-form projection pp , the leading-monomial selector M , and an arbitrary class assignment a are not silently included as definable functions in that structure. Our use of transfer proves bounded-regime statements about L_0 ; it does not claim definability or uniqueness of L_a in an $\mathfrak{o}$ -minimal expansion.

7.3 Local analyticity is not global analytic continuation

The functions L_a have local Hahn–Taylor descriptions: near any point, h_a is a fixed scalar and the remaining Taylor series is the baseline one. This does not provide a classical path joining the point to the real axis through a connected complex domain, nor an identity theorem that propagates an equality between different fine components. A local power-series condition on this topology is therefore intentionally distinguished from a single global transseries prescription.

The distinction is useful for specification as well as for theory. A symbolic system may implement the baseline by an exact normal-form rule. Alternatively it may specify only a recurrence, local derivatives, and asymptotics. These are different interfaces: the second does not determine the first, even after global log-convexity and all finite multiplication formulas are required.

8 Surcomplex log-Gamma and an exact finite-phase tube

8.1 The canonical exponential with finite imaginary part

For a finite surreal $b = r + \epsilon$, where $r \in \mathbb{R}$ and $\epsilon \in \mathfrak{m}$, define

$$\text{cis}(b) = e^{ir} \sum_{k \geq 0} \frac{(i\epsilon)^k}{k!}. \quad (8.1)$$

The sum is strong. Formal exponential identities give $\text{cis}(b + d) = \text{cis}(b) \text{cis}(d)$ for finite b, d and $|\text{cis}(b)| = 1$. Thus

$$E(v + ib) = e^v \text{cis}(b), \quad v \in \mathbf{No}, \quad b \in \mathcal{O}, \quad (8.2)$$

is a canonical additive-to-multiplicative homomorphism on $\mathbf{No} + i\mathcal{O}$. It is finely complex differentiable there, with $E' = E$: on a sufficiently small complex neighborhood the increment has a strong exponential series and $E(w + u) = E(w) \sum u^k/k!$.

No value for $\text{cis}(b)$ with infinite b is prescribed by (8.1). The following construction keeps this domain restriction visible. In particular, an infinite imaginary part of a log-Gamma value is not automatically an argument at which (8.2) may be used.

8.2 The logarithm on the right half-plane

For $z = x + iy$ with $x > 0$, define the right-half-plane logarithm by

$$\text{Log } z = \log|z| + i \text{Arg}(z), \quad -\pi/2 < \text{Arg}(z) < \pi/2.$$

The angle is a finite surreal. It may be constructed without evaluating a trigonometric function at an infinite argument: use the Taylor extension of $\arctan$ at finite y/x , and for infinite $|y/x|$ use $\text{sgn}(y)\pi/2 - \arctan(x/y)$. These formulas agree on their overlaps by real analytic identities and infinitesimal Taylor evaluation. They give $E(\text{Log } z) = z$ and locally $\text{Log}(z + u) = \text{Log } z + \log(1 + u/z)$, with the last logarithm evaluated at a sufficiently small infinitesimal ratio. Hence $\text{Log}'(z) = 1/z$.

Now restrict attention to the fine-open tube

$$\mathcal{T} = \{x + iy \in \mathbf{No}[i] : x > 0, \quad y \in \mathcal{O}\}.$$

If $x > \mathbb{R}$ and y is finite, a particularly simple formula is

$$\text{Log}(x + iy) = \log x + \sum_{k \geq 1} \frac{(-1)^{k+1}}{k} \left(\frac{iy}{x}\right)^k. \quad (8.3)$$

8.3 Definition and functional identities of complex log-Gamma

Let $\ell_{\mathbb{C}}$ be the ordinary holomorphic branch of log-Gamma on the complex right half-plane, chosen to equal $\log \Gamma$ on its positive real axis. At nonzero points of its imaginary boundary, this branch has a local analytic continuation, which is uniquely specified by continuation from the right half-plane.

For finite $z \in \mathcal{T}$ with $\zeta = \text{st}(z) \neq 0$, define $\mathcal{L}_0(z)$ by the complex Taylor series of that local branch at ζ . Here standard part is taken componentwise. For infinitesimal $z \in \mathcal{T}$, define

$$\mathcal{L}_0(z) = -\text{Log } z - \gamma z + \sum_{k \geq 2} \frac{(-1)^k \zeta(k)}{k} z^k. \quad (8.4)$$

For $z = x + iy \in \mathcal{T}$ with $x > \mathbb{R}$, define

$$\mathcal{L}_0(z) = \left(z - \frac{1}{2}\right) \text{Log } z - z + c + \sum_{j \geq 1} b_j z^{-(2j-1)}. \quad (8.5)$$

The inverse z^{-1} is infinitesimal, so this is a well-defined strong sum. Finally set

$$\mathcal{L}_a(x + iy) = \mathcal{L}_0(x + iy) + h_a(x). \quad (8.6)$$

The gauge takes values in $\mathbf{No} \subset \mathbf{No}[i]$, so its imaginary part is zero; its values need not lie in the ordinary real field $\mathbb{R}$.

Theorem 8.1 (Surcomplex logarithmic invariance). *For every assignment a , $\mathcal{L}_a$ is finely holomorphic on $\mathcal{T}$ and has local strongly evaluable Taylor expansions. It restricts to L_a on the positive real-surreal axis, and all its positive-order fine complex derivatives coincide with those of $\mathcal{L}_0$. Moreover,*

$$\mathcal{L}_a(z+1) - \mathcal{L}_a(z) = \text{Log } z, \quad (8.7)$$

$$\mathcal{L}_a(nz) = \sum_{k=0}^{n-1} \mathcal{L}_a(z + k/n) + (nz - \tfrac{1}{2}) \log n - (n-1)c \quad (8.8)$$

for all $z \in \mathcal{T}$ and ordinary $n \geq 1$.

Proof. Local strong summability and differentiability follow exactly as in Proposition 3.1, with complex coefficients. At finite standard part one uses a small ordinary complex disk on which the chosen branch is analytic; at standard part zero use (8.4); at infinite real part use the joint expansions in z^{-1} and a small increment divided by z . The extra term $h_a(\text{Re } z)$ is constant on small fine disks because the real part of a sufficiently small increment is finite. Hence it has zero fine complex derivatives.

For finite z , the classical analytic recurrence and multiplication identities extend by these Taylor expansions. Their logarithmic constants are fixed by agreement on the ordinary positive real axis. The identities remain valid in local disks meeting a nonzero imaginary boundary point by ordinary analytic continuation. Near zero one uses the recurrence-based formula (8.4).

For infinite real part, the formal proofs of Lemma 3.2 and Proposition 3.4 evaluate at z just as at x ; (8.3) ensures consistent logarithmic branches, in particular $\text{Log}(nz) = \log n + \text{Log } z$. The gauge cancels because $h_a(x+1) = h_a(x)$ and $\sum_{k=0}^{n-1} h_a(x + k/n) = h_a(nx)$. $\square$

8.4 The exact phase estimate

The imaginary part of (8.5) grows at the scale $y \log x$, even though y itself is finite. This identifies precisely when the canonical exponential (8.2) is available.

Lemma 8.2 (Imaginary log-Gamma at infinite real part). *Let $x > \mathbb{R}$ and $y \in \mathcal{O}$. Then*

$$\text{Im } \mathcal{L}_0(x + iy) - y \log x \in \mathfrak{m}. \quad (8.9)$$

More precisely, for $y \neq 0$,

$$\frac{\text{Im } \mathcal{L}_0(x + iy)}{y} = \log x - \frac{1}{2x} + O(x^{-2}) + O(y^2 x^{-2}). \quad (8.10)$$

Consequently $\text{Im } \mathcal{L}_a(x + iy)$ is finite if and only if $y \log x$ is finite, for every **No**-valued gauge (8.6).

Proof. Re-expand (8.5) around x in the ratio iy/x . All Taylor coefficients at $x \in \mathbf{No}_{>0}$ lie in **No**, and the imaginary part is the odd Taylor portion:

$$\text{Im } \mathcal{L}_0(x + iy) = yL'_0(x) - \frac{y^3}{3!}L'''_0(x) + \frac{y^5}{5!}L^{(5)}_0(x) - \dots$$

This is an identity of strongly summable joint expansions, not an appeal to a real convergence disk of infinite radius. Formula (3.5) gives $L'_0(x) = \log x - 1/(2x) + O(x^{-2})$. The cubic and higher odd terms, after division by y , have total size $O(y^2 x^{-2})$, since $L'''_0(x) = O(x^{-2})$ and each further odd term gains a factor of order y^2/x^2 . Strong summability justifies the factorization. This proves (8.10); if $y = 0$ the imaginary part is exactly zero. Multiplication by a finite y gives (8.9). Adding $h_a(x) \in \mathbf{No}$ changes no imaginary part. A quantity and an infinitesimally close quantity are finite together. $\square$

Theorem 8.3 (Exact canonical finite-phase Gamma domain). *Let Ω be the domain (1.6). It is fine-open and invariant under $z \mapsto z + 1$, $z \mapsto nz$, and $z \mapsto z + k/n$ for the finite positive indices occurring in the Gamma identities. For every assignment a ,*

$$\mathcal{L}_a(z) \in \mathbf{No} + i\mathcal{O} \iff z \in \Omega \quad (z \in \mathcal{T}).$$

Thus

$$\mathcal{G}_a(z) = E(\mathcal{L}_a(z)) \quad (z \in \Omega) \quad (8.11)$$

is a well-defined finely holomorphic, nowhere-zero surcomplex function. It extends Γ_a and satisfies

$$\mathcal{G}_a(z + 1) = z\mathcal{G}_a(z), \quad (8.12)$$

$$\mathcal{G}_a(nz) = (2\pi)^{(1-n)/2} E((nz - \tfrac{1}{2}) \log n) \prod_{k=0}^{n-1} \mathcal{G}_a(z + k/n). \quad (8.13)$$

Proof. At finite z with nonzero standard part, the imaginary part of $\mathcal{L}_0(z)$ is a real constant plus an infinitesimal. At infinitesimal z , formula (8.4) has finite imaginary part because $\text{Arg } z$ is finite. The gauge has zero imaginary part and changes none of this. At infinite real part, Lemma 8.2 gives exactly the condition defining Ω .

To see openness at an infinite $x + iy \in \Omega$, restrict a perturbation $u + iv$ to $|u| < 1$ and $|v| < 1/\log x$. The new real part is still infinite and $\log(x + u) - \log x = O(u/x)$. Therefore $(y + v) \log(x + u)$ is finite. A further shrink preserves positivity. The finite-real-part portion is already open. For the stated transformations, $\log(nx) = \log x + \log n$ and $\log(x + k/n) - \log x \in \mathfrak{m}$ at infinite x , while all imaginary parts involved remain finite. The defining finite-product condition is therefore preserved.

The complex derivative follows from the local chain rule for strong Taylor expansions and $E' = E$. Nonvanishing follows from $|E(v + ib)| = e^v > 0$. Apply the homomorphism E to (8.7) and (8.8); all arguments have finite imaginary part, including $(nz - \frac{1}{2}) \log n$. Finally $E(\text{Log } z) = z$, giving the claimed identities. $\square$

Example 8.4 (The boundary is a logarithmic scale). At real part $x = \omega$, the point $\omega + i/\log \omega$ lies in Ω and its log-Gamma phase is 1 plus an infinitesimal. The point $\omega + i$ lies in $\mathcal{T}$ but not in Ω , because its log-Gamma phase is $\log \omega$ plus an infinitesimal. On the other hand, a nonzero imaginary increment much smaller than $1/\log \omega$ is permitted. Thus the obstruction is not simply whether y is finite or infinitesimal: it is the product $y \log x$.

Warning 8.5 (An exact domain for one prescription, not a maximality claim). Theorem 8.3 identifies exactly where the canonical *finite-phase exponentiation of this log-Gamma* is defined within $\mathcal{T}$. It does not prove that $\mathcal{G}_a$ has no other extension outside Ω , or that a global exponential cannot be chosen by additional noncanonical phase data. Such choices define a different problem. No value of $\sin \omega$ or $e^{i\omega}$ is selected in this article.

9 What the scalar derivation can distinguish

9.1 The imported differential structure

Let ∂ be the Berarducci–Mantova derivation, normalized by $\partial\omega = 1$. We use its Leibniz rule, strong additivity, exponential compatibility, and constant field $\ker \partial = \mathbb{R}$, all established in [6]. In particular $\partial \log x = \partial x/x$ for $x > 0$. These are properties of scalar surreal numbers. No differentiation of $M(x)$ or $\text{pp}(x)$ as an ordinary coordinate function is assumed.

Proposition 9.1 (The baseline satisfies the scalar chain rule). *For every positive surreal x ,*

$$\partial(L_0(x)) = L'_0(x)\partial x. \quad (9.1)$$

Proof. For a finite appreciable $x = r + \epsilon$, the coefficients of (3.2) are real constants for ∂ , and $\partial\epsilon = \partial x$. Strong additivity and Leibniz give the termwise Taylor chain rule. At positive infinitesimal x , the same argument applies to the series in (3.3), with the logarithmic term handled by $\partial \log x = \partial x/x$. At infinite x , apply Leibniz and logarithmic compatibility to the elementary terms in (3.4), and strong additivity to the Stirling series. All differentiated families are summable by the strong-additivity property of ∂ . The resulting expressions coincide with the fine derivatives computed in Proposition 3.1, multiplied by ∂x . $\square$

9.2 Rigidity of infinitesimal locally constant discrepancies

Theorem 9.2 (Scalar compatibility removes flat local gauges). *Let $h : \mathbf{No}_{>0} \rightarrow \mathbf{No}$ be finely locally constant, zero at positive finite arguments, and infinitesimal at every positive infinite argument. Set $L = L_0 + h$. If*

$$\partial(L(x)) = L'(x)\partial x \quad (x > 0), \quad (9.2)$$

then $h = 0$ and $L = L_0$. In particular this conclusion applies to every algebraically flat gauge constructed in this article.

Proof. Since $h' = 0$, subtract (9.1) from (9.2). It follows that $\partial(h(x)) = 0$, so $h(x) \in \mathbb{R}$ for every x . The only real infinitesimal is zero. At infinite arguments $h(x)$ therefore vanishes, and at finite ones it vanishes by hypothesis. $\square$

The proof uses only infinitesimality of the discrepancy, not its full algebraic flatness. The local-constancy hypothesis is essential to the stated argument: we do not conclude uniqueness among arbitrary functions with a scalar chain rule and a Stirling expansion.

Theorem 9.3 (Complete scalar rigidity inside the monomialwise family). *For the family L_a of (1.1), the scalar chain rule (9.2) holds for all positive surreal arguments if and only if $a = 0$. No convexity or flatness assumption on a is required.*

Proof. The baseline case follows from Proposition 9.1. Conversely, Lemma 5.1 gives $L'_a = L'_0$, so the scalar chain rule forces every $h_a(x)$ to be real. Fix an infinite monomial m and put $r = h_a(m) = ma(m) \in \mathbb{R}$. Since $\sqrt{m}$ is a smaller infinite monomial,

$$h_a(m + \sqrt{m}) = a(m)(m + \sqrt{m}) = r + \frac{r}{\sqrt{m}}.$$

This too must be real. Thus $r/\sqrt{m}$ is real and infinitesimal, hence zero. Therefore $r = 0$ and $a(m) = 0$. As m was arbitrary, $a = 0$. $\square$

Example 9.4 (An explicit derivative mismatch). For $h_\lambda(x)$ from (1.4), its fine coordinate derivative at ω is zero. In contrast, its scalar value there is $\lambda\omega e^{-\omega}$, and

$$\partial(h_\lambda(\omega)) = \lambda(1 - \omega)e^{-\omega}. \quad (9.3)$$

This is nonzero for $\lambda \neq 0$. Consequently

$$\partial(L_\lambda(\omega)) - L'_\lambda(\omega)\partial\omega = \lambda(1 - \omega)e^{-\omega}.$$

The discrepancy is explicit and exactly computed. It is not an assertion that the two notions of derivative are inconsistent; they are different operations and here they fail a particular compatibility requirement.

9.3 Why scalar compatibility alone is not a universal uniqueness axiom

The last theorem is an exact result for its named family. To make its scope sharp, there are other, non-flat gauges which satisfy the scalar chain rule.

Proposition 9.5 (A different real-valued gauge survives scalar compatibility). *For $c_0 \in \mathbb{R}$, define*

$$k_{c_0}(x) = \begin{cases} 0, & x \text{ positive finite,} \\ c_0 \operatorname{st}(x/M(x)), & x \text{ positive infinite.} \end{cases}$$

Then $L_0 + k_{c_0}$ is strictly convex, satisfies the logarithmic recurrence and every finite Gauss formula, has all the same positive-order fine derivatives as L_0 , and satisfies the scalar chain rule. If $c_0 \neq 0$ it is not algebraically flat relative to L_0 .

Proof. The function is real-valued, unchanged by finite translations, and scales by n under $x \mapsto nx$ for positive ordinary integers. Thus all the identity and derivative arguments apply, and $\partial(k_{c_0}(x)) = 0$ because its value is real. For convexity, finite or mixed arguments are treated as before. At infinite arguments the gauge difference is finite. Whenever the baseline gap is positive infinite this is harmless. In the remaining case $y/x \sim 1$, the leading monomial and leading real coefficient of x and y agree, so $k_{c_0}(y) = k_{c_0}(x)$ exactly. Hence every tangent gap remains positive. Finally $k_{c_0}(m) = c_0$ at an infinite monomial m , which is not flat if $c_0 \neq 0$. $\square$

Thus two distinct conclusions should not be confused. Scalar compatibility eliminates every infinitesimal locally constant discrepancy, and it eliminates every monomialwise-linear gauge $a(M(x)) \operatorname{pp}(x)$. It does not by itself eliminate every imaginable finite-valued locally constant discrepancy. Flatness or an equally strong normalization has a real role in a broader uniqueness statement.

10 A reusable convexity principle

The proof is not restricted to the Bernoulli coefficients. It isolates an abstract mechanism useful for other asymptotically convex special functions. We state it with explicit hypotheses rather than claiming unproved extensions to other functions.

Theorem 10.1 (Scale-margin gauge principle). *Let $F : \mathbf{No}_{>0} \rightarrow \mathbf{No}$ be finely differentiable and have strictly positive tangent gaps at distinct arguments. Suppose further that:*

- (a) $D_F(y, x)$ is positive infinite whenever exactly one argument is infinite, or when both are infinite and $y/x \not\sim 1$ (including infinite or infinitesimal ratios);
- (b) if $x \neq y$ are positive infinite and $y/x \sim 1$, then

$$D_F(y, x) \sim \kappa \frac{(y - x)^2}{x}$$

for a fixed positive real κ .

For the gauges h_a in (1.1), $F + h_a$ is convex if and only if $ma(m)$ is finite for every $m \in \mathfrak{M}_\infty$. In the affirmative case it is strictly convex.

Proof. The proof of Theorem 5.4 uses exactly these hypotheses: finite gauge differences are dominated by the gaps in (a), and in (b) the ratio of a nonzero gauge difference to the baseline gap is asymptotic to $|a(m)|x/(\kappa|y-x|)$ when the increment is infinite. It is infinitesimal under the proposed coefficient bound. For finite increments the difference vanishes. If the coefficient bound fails, use the same $\delta = \text{pp } \sqrt{\min(m, |a(m)|m)}$ as in Theorem 5.5; the baseline gap is asymptotic to $\kappa\delta^2/m$ and is dominated by the negative gauge increment. This proves necessity and strict sufficiency. $\square$

For $F = L_0$, Lemmas 4.2 and 4.3 verify all hypotheses with $\kappa = 1/2$. For another function, these hypotheses need separate proofs, especially at extremely small relative increments. Knowing only $F''(x) \sim 2\kappa/x$ does not automatically establish them in the fine surreal calculus. The abstract theorem states precisely what must be checked.

11 Provenance, limitations, and further questions

11.1 Repository overlap audit

The original repository inspection used the tree with observed commit

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and the documentation catalogue [12]. That catalogue listed eighteen research packages, including surreal construction and birthday questions, surcomplex local and global analysis, polynomial algebra, trigonometry, differential equations, rank-one analytic geometry, and spectral theory. The differential-equations article was also inspected to avoid duplicating its scalar-derivation and finite-phase results. This was a targeted catalogue and relevant-source inspection, not a line-by-line audit of every archived manuscript.

No other Gamma-specific research package was identified in that catalogue. This is a historical account of that targeted inspection, not a current exhaustive non-overlap claim. The maintained collection has since grown; the original machine-readable audit in `data/source-audit.json` is retained as a record of the earlier inspection. The present contribution is therefore organized around Gamma uniqueness and its sharp global convexity margin, rather than reproducing the repository's existing general local-constancy, trigonometry, or differential-equation results. The finite-phase exponential and scalar-derivation distinction are background mechanisms; the log-Gamma phase calculation and the exact gauge classification are the Gamma-specific results proved here.

11.2 Published inputs and proposed contributions

The imported components are kept separate from the claims of this article: Conway normal forms and the usual surreal exponential [1, 2]; positive-support Hahn substitution [3]; restricted analytic and exponential transfer [4]; classical Gamma identities, expansions, and remainder bounds [7, 8, 10, 11]; the existing Taylor/transseries extension approach of Costin–Ehrlich [5]; and the scalar derivation of Berarducci–Mantova [6].

The proposed contribution is the combination of the all-scale tangent gap proof, the necessary-and-sufficient bound $ma(m) \in \mathcal{O}$, the flat counterexamples retaining all the stated identities and error inequalities, the exact phase product criterion $y \log x \in \mathcal{O}$, and the scalar rigidity statements

for the specified gauges. The short rigidity deductions and abstract reformulation are consequences supporting the central theorem, not separately advertised solutions of famous open problems.

The literature search included surreal Gamma extensions and Bohr–Møllerup/non-Archimedean uniqueness terminology. The relevant Costin–Ehrlich sections were read, and the classical facts were checked against the NIST Digital Library of Mathematical Functions. I did not locate this exact sharp coefficient theorem and combined invariance result in the sources consulted. Search results for some exact phrases were sparse or noisy. Accordingly, the justified status is *proposed new results with proofs*, not a certificate of priority or an exhaustive negative claim about the entire literature.

11.3 Claims not made

No claim is made that the ordinary Gamma function has been newly constructed, that the original real Bohr–Møllerup theorem is false, or that the Costin–Ehrlich extension is internally ambiguous after its exact extension operator has been fixed. Nor is the coefficient theorem a classification of all surreal Gamma functions. We do not supply a proper-class contour integral, a global surcomplex exponential, or a maximal surcomplex Gamma domain. The asymptotic indices are ordinary integers, and the series are strong sums, not limits of fine-topological partial sums.

The proofs have been algebraically checked in the finite computations described in appendix A, but have not been formalized in Lean or another proof assistant. In particular, computation does not replace the all-scale order comparisons or the class-theoretic support arguments. Independent verification of those proofs and of the scope of the novelty claim remains appropriate before publication.

11.4 Concrete next mathematical questions

The exact family considered here suggests several focused problems, none of which is claimed solved merely by the preceding results.

A full gauge classification. Describe all finite-translation-invariant h for which $L_0 + h$ is convex and $h(nx) = nh(x)$ for every positive ordinary integer. The monomialwise family and the real-valued leading-coefficient family show that there are at least two genuinely different mechanisms. A complete answer would have to describe allowable behavior on $\mathbf{No}/\mathcal{O}$ rather than only on leading monomials.

A minimal uniqueness package. Within locally constant infinitesimal discrepancies, scalar compatibility is sufficient. Can it be replaced by a natural condition stated purely in terms of the function and normal-form truncation? Any candidate must exclude $\lambda \text{pp}(x)e^{-M(x)}$ while preserving the baseline’s local and functional identities. Simply repeating all finite-order asymptotics or all finite Gauss formulas cannot work.

Beyond the finite-imaginary tube. The log-Gamma construction and phase criterion here are restricted to $\mathcal{T}$. Extending a logarithmic prescription through much larger complex sectors requires support control when both components are infinite. Actual Gamma values then additionally require a specified phase structure. These are two separate questions, not one unqualified request to exponentiate an arbitrary surcomplex value.

Proof-assistant decomposition. A realistic formalization can isolate the ordered-field tangent criterion, the algebraic gauge identities, and the abstract scale-margin theorem from the analytic baseline. The first three do not require an implementation of all surreal special functions. The remaining analytic obligations are precisely the strong substitution lemmas, the finite restricted-analytic transfer interface, and the four cases of the infinite tangent-gap lemma. No uncompiled Lean code is presented as a verified formal proof.

12 Conclusion

The positive surreal half-line admits enough separation between finite translations and infinite scales to preserve the entire local Gamma calculus while changing the global function. Global log-convexity does not remove that freedom: its exact tolerance in the monomialwise-linear family is $a(m) = O(m^{-1})$. Exponentially small coefficients lie far inside the permitted region, producing distinct normalized Gamma functions that share all finite multiplication identities, all finite-translation normalized germs, every Stirling coefficient, and every ordinary-order enveloping Stirling inequality.

The phenomenon is not unrestricted arbitrariness. A coefficient larger than the sharp threshold produces an explicit negative tangent gap. The surcomplex logarithmic extension has a precise finite-phase condition. And a scalar chain rule detects discrepancies invisible to fine coordinate derivatives, removing all flat locally constant gauges in the specified category. These three boundaries—convexity scale, phase scale, and scalar compatibility—make the construction a controlled result rather than an isolated pathological example.

A Exact symbolic checks and reproduction

The archive includes `code/verify.py`, a Python program using SymPy, and its recorded JSON output in `data/verification.json`. The recorded run passed all 130 checks (Python 3.13.5 and SymPy 1.14.0; the precise runtime versions are also saved in the JSON record). The calculations use rational arithmetic and formal power series. There are no floating-point tests pretending to represent infinite surreal numbers.

The tests check the Bernoulli normalizations, the shift cancellation underlying (3.9), several finite Gauss defects, the tangent kernel expansions, the quadratic divisibility of (4.11), and finite sparse normal-form identities for pp and M . They also check the displayed derivative mismatch (9.3). The normal-form tests are finite algebraic examples, not a general implementation of the surreal field or its exponential.

The following commands reproduce the checks and build in temporary output directories from the package directory, preserving the recorded evidence:

```
python -m pip install -r data/requirements.txt
python code/verify.py --output /tmp/gamma-verification-rerun.json
mkdir -p /tmp/gamma-article-build
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  -outdir=/tmp/gamma-article-build article.tex
```

Without `latexmk`, run `pdflatex` three times with `-interaction=nonstopmode`, `-halt-on-error`, and `-output-directory=/tmp/gamma-article-build`.

The archived build helper is `code/makefile-Makefile`; its check target overwrites the historical JSON record, so use the explicit temporary output path above for reruns. The LaTeX source has an internal bibliography and no external graphics or custom font files.

A.1 A transparent recurrence calculation

With $t = X^{-1}$ and $R_J(t) = \sum_{j=1}^J b_j t^{2j-1}$, the formal recurrence defect reduces to the ordinary formal expression

$$\left(t^{-1} + \frac{1}{2}\right) \log(1+t) - 1 + R_J\left(\frac{t}{1+t}\right) - R_J(t). \quad (\text{A.1})$$

Its coefficients through degree $2J+1$ vanish. The first coefficient that can be affected by the omitted Stirling term has degree $2J+2$. This finite truncation property is what the program checks. The full proof of Lemma 3.2 establishes cancellation at every ordinary order.

A.2 A transparent multiplication calculation

For $u_k = k/n$, the elementary terms in the Gauss defect reduce, after cancellation of $\log X$, $\log n$, and c , to

$$\sum_{k=0}^{n-1} \left[\left(t^{-1} + u_k - \frac{1}{2}\right) \log(1 + u_k t) - u_k \right].$$

Adding

$$\sum_{k=0}^{n-1} R_J\left(\frac{t}{1 + u_k t}\right) - R_J(t/n)$$

gives the truncated defect tested by the program. Its coefficients through degree $2J$ vanish; an omitted term of degree $2J+1$ can affect the next coefficient. Checking more terms is a useful regression test, but the periodic-series lemma is the proof at all orders.

A.3 What the checks cannot establish

The program does not decide positivity of arbitrary surreal functions, quantify over a proper class, prove Neumann's support lemma, or check first-order transfer. It also does not verify novelty. Its role is to expose coefficient, sign, indexing, or normalization mistakes in the finite algebra supporting the arguments. The actual proof obligations remain visible in the numbered lemmas and theorems.

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